JJBike

1. Objective.

Build a data warehouse so that the administrator of the fictional company JJBike can understand their business.

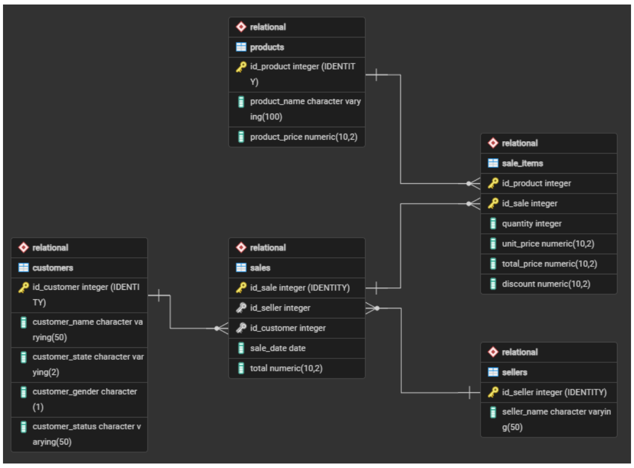
2. Modeling.

To create the analytical database structure model (OLAP) considering the transactional database (OLTP), the subject to be analyzed (the fact), and the dimensions (the various viewpoints from which to analyze the fact (the dimensions)), using the Star Schema model (the most common model for Business Intelligence), the following transformations were performed:

Dimension	Attributes	Source Table
dim_product (what?)	product_key (SK)	-
	id_product	product
	product_name	product
	validity_start_date (versioning)	-
	validity_end_date (versioning)	-
dim_seller (who?)	seller_key (SK)	-
	id_seller	seller
	seller_name	seller
	validity_start_date (versioning)	-
	validity_end_date (versioning)	-
dim_customer (for whom?)	customer_key (SK)	-
	id_customer	customer
	customer_name	customer
	customer_state	customer
	customer_gender	customer
	customer_status	customer
	validity_start_date (versioning)	-
	validity_end_date (versioning)	-
dim_time (when?)	time_key (SK)	-

	date	-
	time_day	-
	time_month	-
	time_year	-
	time_weekday	-
	time_quarter	-

Fact	Consolidated Table	Granularity	Attributes	Source Table
fact_sales	sales	Item sold.	sale_key (SK)	-
			seller_key (FK)	dim_seller
			customer_key (FK)	dim_customer
			product_key	dim_product
			time_key (FK)	dim_time
			quantity	sale_items
			unit_price	sale_items
			total_price	sale_items
			discount	sale_items



3. Transactional environment (OLTP) - Microsoft SQL Server Management Studio (SSMS).

3.1. Creation of the relational schema.

The Relational database was created, serving as the source layer for the transactional data of the JJBike company. Within it, the file 1. Scripts_Create_Table_Relational.sql created five operational tables. In all of them, the primary key is automatically generated by the IDENTITY clause, eliminating the need to control external sequences. All columns, except the primary keys themselves, were defined with NOT NULL, ensuring that no record is persisted with mandatory data open.

- **sellers:** Stores the sellers' data. The seller_name attribute identifies the seller's name and is the only field besides the PK, reflecting the intentional scope of this entity in the transactional model;
- **products:** Stores the product catalog. The product_name attribute identifies the product and product_price records the current list price at the time of registration. Both are mandatory, as a product without a name or price cannot participate in a sales process; customers: Stores customer registration information. The customer_name attribute identifies the customer; customer_state records the state of origin (Varchar(2)); customer_gender records the gender in a single character (Char(1)); and customer_status records the customer's plan level, an attribute that, because it changes over time, will be the main element tracked by SCD Type 2 in the analytical environment;
- **sales:** Represents the header of each sale. The id_seller attribute is a foreign key that references Relational..sellers, and id_customer is a foreign key that references Relational..customers, both mandatory, as a sale cannot exist without an associated seller and customer. The sale_date attribute records the date on which the transaction occurred with type DATETIME, and total stores the consolidated value of the sale;
- **sale_items:** Linking table between sales and products, with a primary key composed of id_product and id_sale. Referential integrity constraints were declared via ALTER TABLE after table creation:
    - FK_SALEITEMS_PRODUCT in id_product: references Relational..products, preventing historical data from losing the reference to the transacted product;
    - FK_SALEITEMS_SALE in id_sale: references Relational..sales, ensuring that items are always associated with an existing sale;
    - The attributes quantity, unit_price, total_price, and discount are the detail metrics for each item and were declared NOT NULL because they will directly populate the measure columns of the fact table in the analytical environment.

3.2. Table Population.

The data was inserted using the files 2. Insert_Into_Customers.sql, 3. Insert_Into_Products.sql, 4. Insert_Into_Sellers.sql, 5. Insert_Into_Sales.sql, and 6. Insert_Into_SaleItems.sql. Each file executes INSERT INTO commands on the respective table in the Relational database, specifying only the data columns, primary keys are omitted because the IDENTITY mechanism assigns them automatically and sequentially to each inserted record.

4. Staging environment (invisible/conceptual).

There was no treatment or cleaning of raw data before creating the dimensions; the direct data transfer between PostgreSQL schemas itself acted as the load trigger.

5. Analytical environment (OLAP) - Microsoft SQL Server Management Studio (SSMS).

5.1. Creation of the database and dimensional tables.

The Dimensional database was created, representing the Data Warehouse structured in a Star Schema model. The file 1. Scripts_Create_Table_Dimensional.sql created the dimensions dim_seller, dim_customer, dim_product, and dim_time, in addition to the fact table fact_sales. In all tables, the Surrogate Key (SK) is automatically generated by IDENTITY, serving as the internal primary key of the DW, decoupled from the source IDs of the transactional system.

All columns were defined with NOT NULL, with the sole exception of validity_end_date in dimensions with SCD Type 2. This field remains NULL while the record is active; it is only filled with the current time when an attribute change is detected and a new version of the record needs to be created. The validity fields are of type DATETIME, allowing the differentiation of records processed on the same date with millisecond precision.

Dimensional tables have the following attribute structure:

- **dim_seller:** seller_key is the automatically generated SK and acts as the internal PK of the DW. id_seller preserves the source ID from Relational..sellers, maintaining traceability between the two environments. seller_name carries the seller's name. validity_start_date receives GETDATE() when the record is inserted, marking the start of its validity. validity_end_date remains NULL while the record is the active version;
- **dim_customer:** customer_key is the SK, the internal PK of the DW. id_customer preserves the source ID. customer_name, customer_state, customer_gender, and customer_status are the tracked descriptive attributes, any change in any of them triggers SCD Type 2 versioning. validity_start_date and validity_end_date control the validity period of each version of the record;
- **dim_product:** product_key is the SK. id_product preserves the source ID. product_name is the only descriptive attribute tracked, a change in its value triggers versioning. validity_start_date and validity_end_date function the same way as in other SCD Type 2 dimensions;
- **dim_time:** time_key is the SK. time_date stores the complete date and serves as a linking field with the transactional system during fact loading. time_day, time_month, time_year, and time_quarter are derived from time_date and enable analytical groupings by temporal granularity. time_weekday stores the day of the week as an integer (1 = Sunday, 7 = Saturday), following the SQL Server DATEPART(weekday, ...) pattern. This dimension does not have validity fields because time is immutable;
- **fact_sales:** sale_key is the SK of the fact. seller_key, customer_key, product_key, and time_key are the foreign keys (FKs) that reference their respective dimensions, all declared NOT NULL, since each fact must be associated with all dimensions of the model. quantity, unit_price, total_price, and discount are the metrics for each item sold, also NOT NULL.

5.2. Populating the time dimension.

The dim_time dimension was populated by the file 2. Insert_Into_dim_time.sql. The strategy adopted was the programmatic generation of a continuous date range from January 1, 1970 to December 31, 2030. The script consists of three chained blocks:

- **Date format configuration:** SET DATEFORMAT dmy defines the interpretation of dates in the day/month/year pattern for the session, ensuring that literal dates are read correctly by SQL Server;

- **Recursive CTE date_sequence:** uses a CTE anchored at @start_date and recurses via DATEADD(day, 1, generated_date) until it reaches @end_date. The OPTION (MAXRECURSION 0) clause removes the default 100 iteration limit of SQL Server, allowing the recursion to traverse all days of the interval without interruption;
- **INSERT final:** inserts each generated date into dim_time, extracting time_day via DAY(), time_month via MONTH(), time_year via YEAR(), time_weekday via DATEPART(weekday, ...) and time_quarter via DATEPART(quarter, ...).

5.3. Extraction (E) and loading (L) of data from the transactional environment (OLTP) to the analytical environment (OLAP).

Data loads were performed using the updated 3. Script.sql file, structured in sequential phases to demonstrate the comprehensive operation of SCD Type 2 along with advanced historical and referential integrity mechanisms.

5.3.1. Initial Loading of Dimensions.

The loading pattern for the three dimensions supporting SCD Type 2 was enhanced to elegantly handle both attribute mutations and physical source deletions (*Hard Deletes*). Before executing each load block, two temporal control variables are evaluated: @dt_end captures the precise instant of execution via GETDATE() and is used to expire old or deleted records; @dt_start adds 3 milliseconds to @dt_end via DATEADD(MILLISECOND, 3, @dt_end) and establishes the validity_start_date for newly generated versions. The mechanism runs through two independent steps:

- **Deletion scanning:** A standalone UPDATE statement executes before the merge process, probing the dimension to expire (SET validity_end_date = @dt_end) any currently active rows (validity_end_date IS NULL) whose business keys (IDs) can no longer be found within the Relational source tables;
- **MERGE and External INSERT processing:** The core block executes a MERGE statement, aligning the source with target records that are currently active (T.validity_end_date IS NULL). When active attributes differ (WHEN MATCHED AND), the target record is expired using @dt_end. If the ID does not exist in the target warehouse (WHEN NOT MATCHED BY TARGET), a clean record is created starting at @dt_start. The OUTPUT $ACTION clause channels mutated rows into an outer INSERT subquery, which isolates action = 'UPDATE' to inject the new active historical version into the dimension with validity_start_date = @dt_start and a null end date.

5.3.2. Loading the fact table (month by month).

The fact_sales table is populated via an INSERT INTO ... SELECT query that consolidates raw operational events with the correct historical state of the dimensions. Sales records were processed incrementally by month (January, February, March, and subsequent months) to evaluate SCD Type 2 interactions over time.

To prevent the critical data warehousing flaw known as "blurred facts"—where late re-runs or retrofitted loads mistakenly align past events with current dimension states—the script implements robust temporal joins instead of blind IS NULL filters:

- **Metrics:** quantity, unit_price, total_price, and discount are mapped directly from Relational..sale_items to preserve the atomic granularity of each line item;
- **Surrogate Keys:** seller_key, customer_key, and product_key are resolved by joining dimensions against the point-in-time date of the sale (v.sale_date). The join logic ensures that the sale falls precisely within a version's historical validity window: v.sale_date >= validity_start_date AND (v.sale_date <= validity_end_date OR validity_end_date IS NULL). This locks each event to the exact profile the entity possessed when the transaction occurred;
- **time_key:** Resolved by matching v.sale_date to the static time_date column in the dim_time calendar dimension;
- **Temporal filter:** The MONTH(v.sale_date) expression restricts data extraction to the exact simulation slice (= 1 for January, = 2 for February, = 3 for March, and > 3 for the remaining period).

5.3.3. Simulation of status change and history processing.

To demonstrate chronological slicing, two distinct operational adjustments were pushed directly onto the Relational..customers source table:

- **Following the January load:** Customers with IDs 1 through 5 were updated to a 'Gold' customer_status;
- **Following the March load:** The customer with ID 3 was promoted to a 'Platinum' customer_status.

Following each operational mutation, the full dimension lifecycle routine (deletion sweep + MERGE with OUTPUT) was executed. The ETL layer isolated the customer_status deviation, successfully stamped an expiration date onto the active version's surrogate key, and generated a brand-new timeline slice starting at @dt_start.

5.3.4. Verification and Audit of SCD Type 2.

Analytical audit queries crossing fact_sales, dim_customer, and dim_time via their surrogate keys verified the structural health of the schema:

- January sales records point correctly to the original SKs of customers 1 to 5, accurately locking in their pre-Gold status portrait;
- February transactions seamlessly index the intermediate 'Gold' SKs produced after the initial batch change;
- A query explicitly filtering for t.time_month = 3 establishes that none of the monitored customer IDs made purchases in March, confirming expected business behavior;
- The final comprehensive audit run after ingestion of the remaining months (> 3) displays a flawlessly balanced data warehouse: older sales stay tethered to historical states (original or 'Gold' SKs), while late sales tie directly to customer 3's latest 'Platinum' SK, proving full temporal consistency.

5.4. KPI (Key Performance Indicator).

To measure JJBike's revenue against monthly targets, the Dimensional..kpi table was created using the 4.KPI.sql file. The script consists of four main structures:

- **Preliminary cleanup:** the IF OBJECT_ID('Dimensional..kpi', 'U') IS NOT NULL DROP TABLE Dimensional..kpi block checks if the table already exists before recreating it, avoiding errors in script re-executions;
- **SELECT block:** defines the three columns that will make up the table: dt.time_month aliased as month, which identifies the reference month; SUM(fs.total_price) aliased as total_revenue, which aggregates the revenue actually earned by summing the total_price of all fact_sales records for the respective month; and a CASE dt.time_month block that assigns a fixed target value for each month, from R$ 220,000 in January to R$ 270,000 in December, aliased as target;
- **Type conversion:** the CAST(... AS NUMERIC(18,2)) operator positioned after the END of the CASE ensures that the target column is consolidated with the same fixed-precision numeric type as total_price in fact_sales. Without this conversion, SQL Server would treat the CASE values as integers, causing type incompatibility when calculating achievement percentages;
- The FROM block starts with Dimensional..fact_sales and performs an INNER JOIN with Dimensional..dim_time based on the condition fs.time_key = dt.time_key, relating each fact record to its respective month. The GROUP BY dt.time_month consolidates all sales for each month into a single row, and the ORDER BY dt.time_month ASC organizes the result in ascending chronological order. The SELECT INTO operator physically creates the Dimensional..kpi table from the query result.

6. Artifacts - Power BI.

The tables from the Dimensional database, created in SSMS, were connected to Power BI.

6.1. Reports.

6.1.1. Transformations (T) in the tables.

In fact_sales, a column called discount_percent (with two decimal places) was created with the following formula:

- discount_percent = TRUNC(('dimensional fact_sales'[discount] / 'dimensional fact_sales'[total_price]) * 100, 2);

In dim_customer, a column called location_map was created with the following formula:

- location_map =
  ```
  SWITCH(
      'dimensional dim_customer'[customer_state],
      "AC", "Acre",
      "AL", "Alagoas",
      "AM", "Amazonas",
      "AP", "Amapa",
      "BA", "Bahia",
      "CE", "Ceara",
      "DF", "Federal District",
      "ES", "Espirito Santo",
      "GO", "Goias",
  ```

"MA", "Maranhao",
"MG", "Minas Gerais",
"MS", "Mato Grosso do Sul",
"MT", "Mato Grosso",
"PA", "Para",
"PB", "Paraiba",
"PE", "Pernambuco",
"PI", "Piaui",
"PR", "Parana",
"RJ", "Rio de Janeiro",
"RN", "Rio Grande do Norte",
"RO", "Rondonia",
"RR", "Roraima",
"RS", "Rio Grande do Sul",
"SC", "Santa Catarina",
"SE", "Sergipe",
"SP", "Sao Paulo",
"TO", "Tocantins",
"Unknown"
) & ", Brazil".

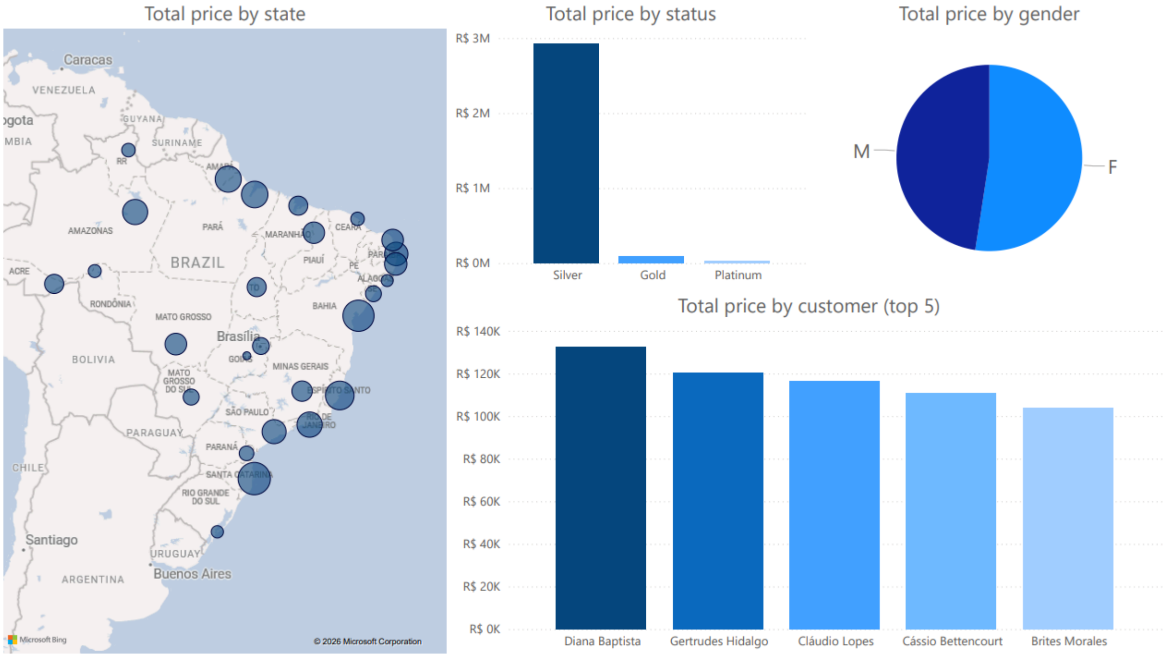
6.1.2. Metrics.

In dim_kpi, three metrics were created so that, in the KPI report, the card has dynamic behavior according to what is selected in the button slicer:

- measure_target =
  IF(
  ISFILTERED('dimensional kpi'[month]) || ISFILTERED('dimensional dim_time'[time_year]),
  SUM('dimensional kpi'[target]),
  0
).
- measure_total_revenue =
  IF(
  ISFILTERED('dimensional kpi'[month]) || ISFILTERED('dimensional dim_time'[time_year]),
  SUM('dimensional kpi'[total_revenue]),
  0
).

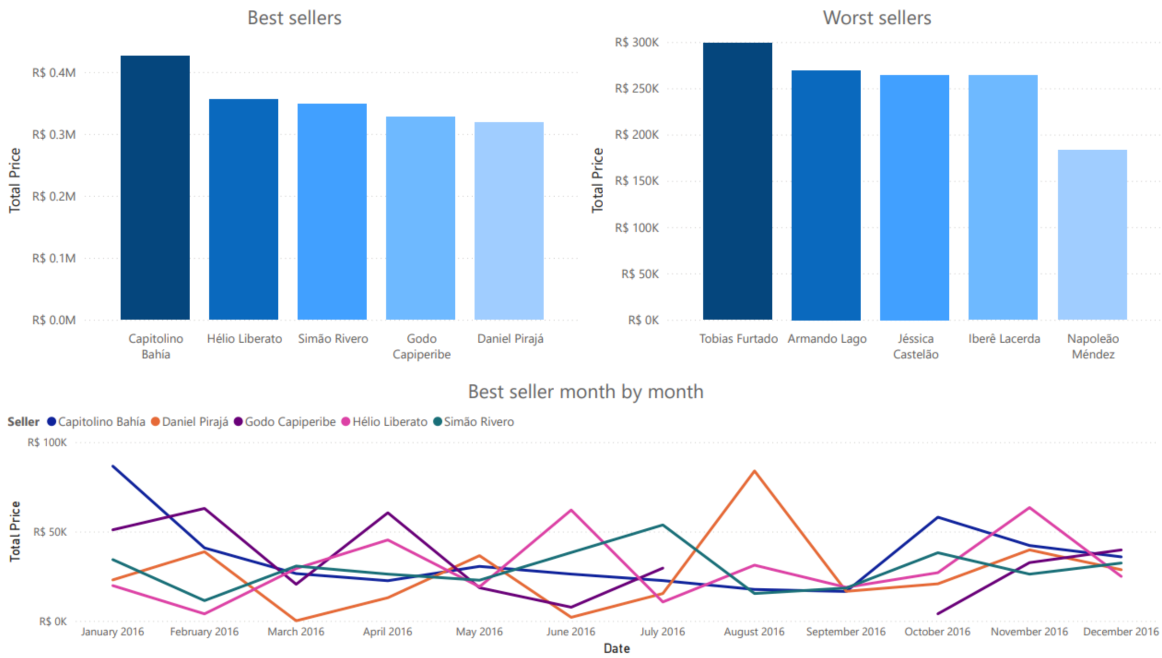
6.1.3. Created Reports.

(Customers)
The customer report shows: total value sum by state, total value sum by status, total value sum by gender, and total value sum by customer (top 5).



(Sellers)

The sellers report shows: total value sum by sellers (top 5 best sellers, top 5 worst sellers, and top 5 best sellers compared month by month).



(Products)

The product report shows: total value per product (top 5 best-selling products and top 5 least-selling products) and total discount per product (top 5 products with the highest accumulated discount and top 5 products with the lowest accumulated discount).

6.2. Dashboards.

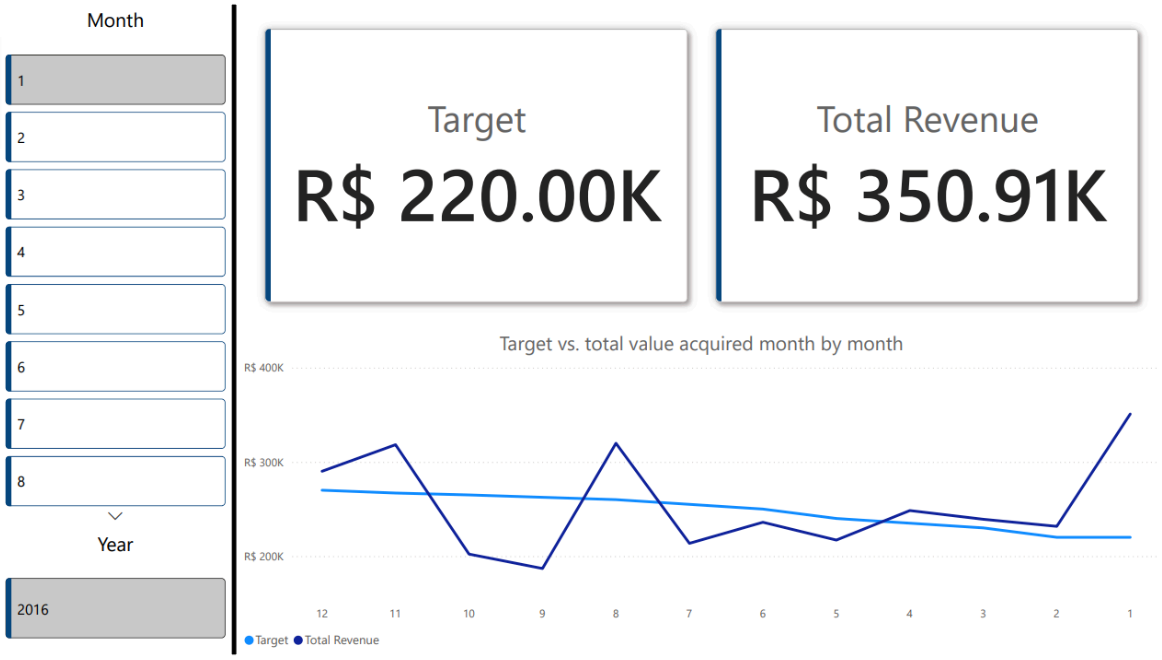
The following dashboards were created:

(Sales)

The sales report shows: total value of products sold (compared month by month), total quantity of products sold (compared month by month), total discount of products sold (compared month by month), and product, sellers, and customer cards for filtering.

(KPI)

The KPI report shows: target, total value acquired, sum of target and total value acquired (compared month by month), and month and year cards for filtering.



Generated file is available in '2. Artifacts/1. Reports+Dashboards/'.

7. Artifact - Microsoft Visual Studio.

7.1. Cube.

The Dimensional database tables were structured and hosted on a full instance of SQL Server (Developer Edition), ensuring native support for Business Intelligence (BI) functionalities;

A Data Source View (DSV) was created, a visual environment where the fact and dimension tables were mapped (excluding the kpi table), generating a logical model in Star Schema format;

The following dimensions were created:

- **dim_customer:** customer_key, customer_name, customer_gender, customer_state, and customer_status;
- **dim_product:** product_key and product_name;
- **dim_time:** time_key, time_year, time_quarter, time_month, and time_day;
- **dim_seller:** seller_key and seller_name.

The following hierarchies were created from the dimensions:

- **Geographic Hierarchy:** customer_state → customer_name (dim_customer).
  - **Note:** In the Attribute Relationships tab, the order should be: Customer Key → Customer Name → Customer State.
- **Temporal Hierarchy:** time_year → time_quarter → time_month → time_day (dim_time).
  - **Note:** In the Attribute Relationships tab, the order should be: Time Key → Time Day → Time Month → Time Quarter → Time Year.
- **Observations:**

- customer_gender and customer_status (from dim_customer) should remain free in the side panel as Non-hierarchical attributes, as inserting them within Geographic Hierarchy would imply that they are part of a geographic subdivision;
- dim_seller and dim_product have only one attribute in addition to their keys, therefore they did not generate hierarchies, and their attributes became Non-hierarchical attributes;
- To avoid duplicate key errors in the Analysis Services engine during dim_time processing, the KeyColumns properties were customized as composite keys, linking time_quarter with time_year, time_month with time_year, and time_day with time_month and time_year, ensuring the uniqueness and integrity of the attribute relationship tree.

The cube was structured from the fact_sales fact table. During the process, relationship keys and analytically incorrect fields (such as unit_price, whose aggregation by sum makes no business sense) were ignored. Only the actual performance metrics were retained (quantity, total_price, discount, and automatic record counting);

A successful deployment of the project was performed on the local instance of the Analysis Services server, compiling the structures and loading the data into memory;

Visual representation of the created cube:

Customer State	Customer Name	Time Year	Time Quarter	Time Month	Time Day	Quantity	Discount	Total Price
AC	Alarico Quinterno	2016	4	12	10	1	60.45	155
AC	Antônio Sobral	2016	3	9	17	1	1593.36	8852
AC	Antônio Sobral	2016	4	10	16	2	1741.25	7935.6
AC	Basilio Soares	2016	1	3	29	1	4.65	155
AC	Basilio Soares	2016	2	4	17	9	879.08	22167.65
AC	Basilio Soares	2016	2	4	18	3	125.1	3289
AC	Basilio Soares	2016	2	5	14	1	64.07	2135.52
AC	Basilio Soares	2016	2	6	6	1	650.93	6509.3
AC	Basilio Soares	2016	2	6	24	1	16.92	188
AC	Basilio Soares	2016	3	7	5	2	775.25	7793
AC	Carla Briones	2016	4	12	5	1	587.75	2260.58
AC	Ester Castanho	2016	3	9	26	5	2327.33	15416.98
AC	Evaristo Bahía	2016	4	10	10	3	692.17	4836.7
AC	Irene Villanueva	2016	4	11	12	3	1654.34	7183.75
AL	Cid Pardo	2016	3	8	21	3	1769.02	13295
AL	Cid Pardo	2016	4	10	18	3	2430.31	13522.61
AM	Deise Farias	2016	3	8	22	3	1231.04	8976.8
AM	Deise Farias	2016	3	9	7	1	1472.16	9201
AM	Diana Baptista	2016	1	2	12	4	79.67	11112.58
AM	Diana Baptista	2016	1	2	13	1	1.69	169.2
AM	Diana Baptista	2016	1	3	15	4	285.75	7205.75
AM	Diana Baptista	2016	2	4	2	1	56.31	5631.01

Generated files are available in '2. Artifacts/2. Cube/'.